

NWIS

DAILY DRILLING REPORT

Doc ID
W001-DDR-14

Well: Alpha-01 | Well ID: W001 | Field: Field_Alpha | Date: 15-05-2024

PRIMARY ISSUE CATEGORY: MUD LOSS

1. REPORT SUMMARY

Report No.	14	Date	15-05-2024
Start Depth	2880 m	End Depth	2910 m
Progress (24 hr)	30 m	Formation	Formation_X (sand)
Rig	Rig Sagar-3	Mud Type	Water-Based Mud (WBM) - KCl/Polymer

2. MUD PROPERTIES

MW (SG)	PV (cP)	YP (lb/100ft2)	Gel 10s/10m	pH	Solids (%)
1.22 -> 1.18	24	18	8/22	9.8	6.5

3. DRILLING PARAMETERS

Depth (m)	ROP (m/hr)	WOB (klb)	RPM	MW (SG)	Torque (kft-lb)
2880	22	11	105	1.22	9.5
2890	20	12	108	1.22	9.8
2900	15	12	110	1.20	10.1
2905	6	10	95	1.18	8.4
2910	17	12	110	1.18	9.6

4. MUD LOSS VOLUME LOG

Time	Pit Volume (bbl)	Loss Rate (bbl/hr)	Action
06:00	620	-	Drilling ahead, normal returns
06:20	590	~8	Loss detected, pumps slowed
06:35	575	~6	LCM pill mixing started
06:55	568	~2	LCM pill pumped and spotted
07:45	566	<1	Losses controlled, MW reduced to 1.18 SG
08:15	566	0	Full returns confirmed

5. OPERATIONS TIMELINE (24 HR)

Time	Duration (hr)	Activity
00:00-06:00	6.0	Drilling ahead 2880-2900 m

06:00-08:00	2.0	Partial mud loss detected at 2905 m; drilling suspended
08:00-11:00	3.0	LCM pill mixed and pumped; mud weight adjusted
11:00-12:30	1.5	Circulated bottoms-up; confirmed loss control
12:30-24:00	11.5	Drilling ahead 2905-2910 m; monitoring returns

6. EVENT / INCIDENT DETAIL

At 2905 m, partial mud loss was detected while drilling through the Formation_X high-permeability sand streak; pit volume dropped ~40 bbl.

Action Taken:

- 1. Drilling was stopped and the string was pulled clear of the bit.
- 2. Pump rate reduced to 40 SPM to lower ECD immediately.
- 3. 35 bbl LCM pill (60 lb/bbl CaCO3 + mica blend) mixed and pumped across 2895-2915 m.
- 4. Mud weight was reduced from 1.22 SG to 1.18 SG.
- 5. Circulation was monitored for 90 minutes to confirm return of full returns.

Result: Mud losses reduced from approximately 8 bbl/hr to negligible levels. Pit volume stabilized at 566 bbl. Drilling resumed at 2907 m.

NPT Incurred: 6.5 hours

Engineer Remark: Third occurrence of this loss pattern in Field_Alpha (see W003 for a comparable event). Recommend flagging 2880-2950 m as a standing loss-risk interval in the NWIS early-warning model.

7. PERSONNEL ON SITE

Company Man	R. Verma	Drilling Supervisor	A. Khan
Mud Engineer	S. Iyer	Toolpusher	M. Das

This Daily Drilling Report is a synthetically generated historical record produced for the NWIS prototype (W001). Not associated with an actual drilling operation.